



Security Operations Center (SOC) Fundamentals and Security Monitoring

1.Objective

The objective of this task is to understand the fundamental concepts of a Security Operations Center (SOC), security monitoring, log management, incident response, and security tools. The task also focuses on gaining practical experience in log analysis, dashboard creation, alert configuration, and security event documentation.

2.Introduction

A Security Operations Center (SOC) is a centralized team responsible for continuously monitoring, detecting, investigating, and responding to cybersecurity threats. SOC analysts use various security tools such as SIEM, IDS/IPS, EDR, and vulnerability scanners to identify malicious activities and protect organizational assets.

The SOC operates 24/7 to ensure timely detection and response to security incidents while maintaining the confidentiality, integrity, and availability of information systems.

3. Theory

3.1 SOC Fundamentals and Operations

A SOC is responsible for:

- Continuous monitoring of network and system activities.
- Detecting suspicious behavior and security incidents.
- Investigating alerts generated by security tools.
- Responding to threats and minimizing impact.

SOC Roles

Tier 1 Analyst

- Monitors alerts.
- Performs initial investigation.
- Escalates incidents when necessary.

Tier 2 Analyst

- Conducts deeper investigations.
- Performs threat analysis.
- Handles incident response activities.



Tier 3 Analyst / Threat Hunter

- Performs advanced threat hunting.
- Develops detection rules.
- Investigates sophisticated attacks.

SOC Manager

- Oversees SOC operations.
- Manages analysts and resources.
- Ensures compliance with security policies.

3.2 Security Monitoring Basics

Security monitoring involves the continuous observation of systems, applications, and network traffic to detect security threats.

Objectives

- Detect unauthorized access attempts.
- Identify malicious activities.
- Monitor policy violations.
- Improve incident response capabilities.

Common Tools

- SIEM (Splunk, Elastic SIEM, Wazuh)
- Network Monitoring Tools (Wireshark)
- IDS/IPS (Snort)
- EDR Solutions (CrowdStrike)

Key Metrics

- Mean Time to Detect (MTTD)
- Mean Time to Respond (MTTR)
- False Positive Rate
- False Negative Rate

3.3 Log Management Fundamentals

Log management consists of:

1. Log Collection
2. Log Normalization
3. Log Storage
4. Log Retention
5. Log Analysis



Common Log Sources

- Windows Event Logs
- Syslog
- Web Server Logs
- Firewall Logs
- Application Logs

Example KQL Query

source = "security-login-"

EventID = 4625

| stats count by SourceIP

This query identifies failed login attempts and groups them by source IP address.

3.4 Security Tools Overview

Tool	Purpose
Splunk	Security Information and Event Management
Wazuh	Open-source XDR and SIEM
Wireshark	Network Packet Analysis
Snort	Intrusion Detection and Prevention
Nessus	Vulnerability Assessment
Osquery	Endpoint Monitoring

Example Snort Rule

```
alert tcp any any -> any 80 (msg:"Malicious Domain";  
content:"malicious.com";  
http_uri;  
sid:1000001;)
```

This rule generates an alert whenever a user attempts to access the specified malicious domain.

3.5 Basic Security Concepts

CIA Triad

Confidentiality

Protects data from unauthorized access.

Integrity

Ensures data remains accurate and unaltered.

Availability

Ensures systems and data remain accessible.



Threat, Vulnerability, and Risk

- Threat: Potential danger to an asset.
- Vulnerability: Weakness that can be exploited.
- Risk: Likelihood of a threat exploiting a vulnerability.

Defense in Depth

Uses multiple layers of security controls to protect systems.

Zero Trust

Never trust, always verify.

3.6 Security Operations Workflow

The SOC workflow consists of:

Detection

Security tools generate alerts.

Triage

Analysts prioritize alerts based on severity.

Investigation

Analysts analyze logs and indicators of compromise.

Response

Threats are contained and eradicated.

Recovery

Systems are restored to normal operation.

3.7 Incident Response Lifecycle

The incident response lifecycle includes:

1. Preparation
2. Identification
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

This structured approach ensures effective handling of security incidents.



3.8 Documentation Standards

SOC teams maintain:

- Incident Reports
- Standard Operating Procedures (SOPs)
- Runbooks
- Post-Incident Reviews

Documentation helps maintain consistency and supports future investigations.

4. Practical Activities

4.1 Log Analysis Practice

Linux Log Analysis

Log analysis was performed using Linux system logs generated through the logger command and reviewed using journalctl and authentication logs.

Event ID	Description
4625	Failed Login Attempt
7045	New Service Installation

The logs were filtered and reviewed to identify suspicious activities and possible brute-force attacks.

```
kali@kali: ~$ cat /home/kali/.zsh_history
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali) ~$
(kali@kali) ~$ logger "soc test a message"
(kali@kali) ~$
(kali@kali) ~$ sudo journalctl -n 20
[sudo] password for kali:
May 29 13:00:01 kali CRON[46367]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
May 29 13:00:01 kali CRON[46369]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi)
May 29 13:00:01 kali CRON[46367]: pam_unix(cron:session): session closed for user root
May 29 13:00:04 kali systemd[1]: Starting phpsessionclean.service - Clean php session files ...
May 29 13:00:04 kali systemd[1]: phpsessionclean.service: Deactivated successfully.
May 29 13:00:04 kali systemd[1]: Finished phpsessionclean.service - Clean php session files.
May 29 13:12:10 kali kernel: e1000 0000:00:03:0 eth0: entered promiscuous mode
May 29 13:15:01 kali CRON[49427]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
May 29 13:15:01 kali CRON[49429]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
May 29 13:15:01 kali CRON[49427]: pam_unix(cron:session): session closed for user root
May 29 13:15:20 kali kali[49598]: soc test a message
May 29 13:16:12 kali dbus-daemon[1101]: [session uid=1000 pid=1101 pidfd=5] Activating via systemd: service name='org.xfce.Xfconf' unit='Xfconfd.service' requested by user 1000
May 29 13:16:12 kali systemd[1079]: Starting xfconfd.service - Xfce configuration service ...
May 29 13:16:12 kali dbus-daemon[1101]: [session uid=1000 pid=1101 pidfd=5] Successfully activated service 'org.xfce.Xfconf'
May 29 13:16:12 kali systemd[1079]: Started xfconfd.service - Xfce configuration service.
May 29 13:17:00 kali sudo[50448]:      kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -n 20
May 29 13:17:00 kali sudo[50448]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
May 29 13:17:01 kali CRON[50494]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
May 29 13:17:01 kali CRON[50496]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
May 29 13:17:01 kali CRON[50494]: pam_unix(cron:session): session closed for user root
lines 1-20/20 (END)
```



Failed Login Analysis

```
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ su root
Password:
su: Authentication failure
kali@kali:~$
```

4.2 Security Event Documentation

A standard event template was created.

Date/Time	Source IP	Event ID	Description	Action Taken
28-May-2026 10:15 AM	192.168.1.10	4625	Multiple Failed Logins	Investigated and Monitored

4.3 Monitoring Dashboard Setup

A dashboard was configured using SIEM tools to display:

- Top Source IP Addresses
- Failed Login Events
- Critical Security Alerts
- Event Trends

The dashboard improved visibility into security events and network activities.

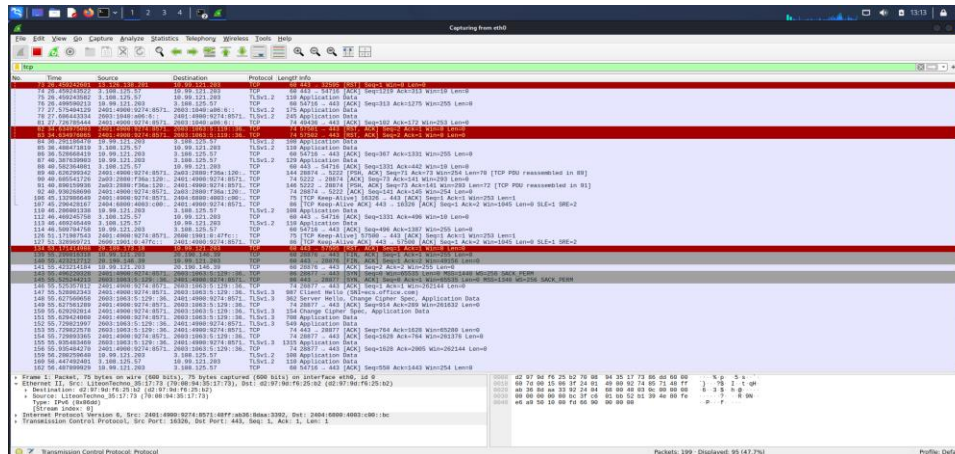


Figure: TCP packet analysis using Wireshark.

ICMP Traffic Analysis

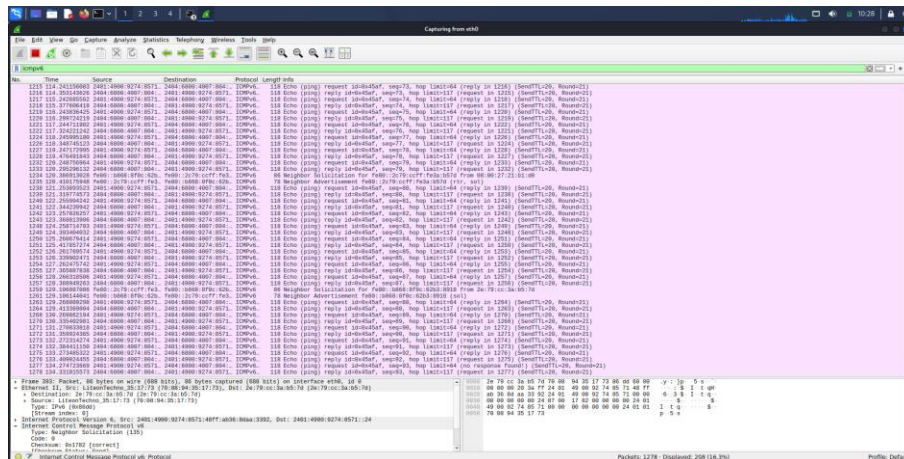


Figure: ICMP packets captured using Wireshark.

DNS Traffic Analysis

Figure: DNS query and response traffic captured using Wireshark.

4.4 Alert Rule Configuration

A custom rule was designed to detect multiple failed login attempts.

Rule Configuration

- Index: security-login-*
- Condition: More than 5 failed logins within 5 minutes

```

kali@kali:~$
┌───( Actions | Edit | View | Help )
May 29 13:09:04 kali systemd[1]: phpsessionclean.service: Deactivated successfully.
May 29 13:09:04 kali systemd[1]: Finished phpsessionclean.service - Clean php session files.
May 29 13:12:10 kali kernel: e1000 0000:00:03:0 eth0: entered promiscuous mode
May 29 13:15:01 kali CRON[49427]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
May 29 13:15:01 kali CRON[49429]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1)
May 29 13:15:01 kali CRON[49427]: pam_unix(cron:session): session closed for user root
May 29 13:15:20 kali kali[49598]: soc test a message
May 29 13:16:12 kali dbus-daemon[1101]: [session uid=1000 pid=1101 pidfd=5] Activating via systemd: service name='org.xfce.Xfconf' unit='xfconfd.service' requested by uid=1000
May 29 13:16:12 kali systemd[137]: Starting xfconfd.service - Xfce configuration service...
May 29 13:16:12 kali dbus-daemon[1101]: [session uid=1000 pid=1101 pidfd=5] Successfully activated service 'org.xfce.Xfconf'
May 29 13:16:12 kali systemd[1079]: Started xfconfd.service - Xfce configuration service.
May 29 13:17:00 kali sudo[50448]:      kali : TTYpts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journaldctl -n 20
May 29 13:17:00 kali sudo[50448]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
May 29 13:17:01 kali CRON[50494]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
May 29 13:17:01 kali CRON[50496]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
May 29 13:17:01 kali CRON[50494]: pam_unix(cron:session): session closed for user root

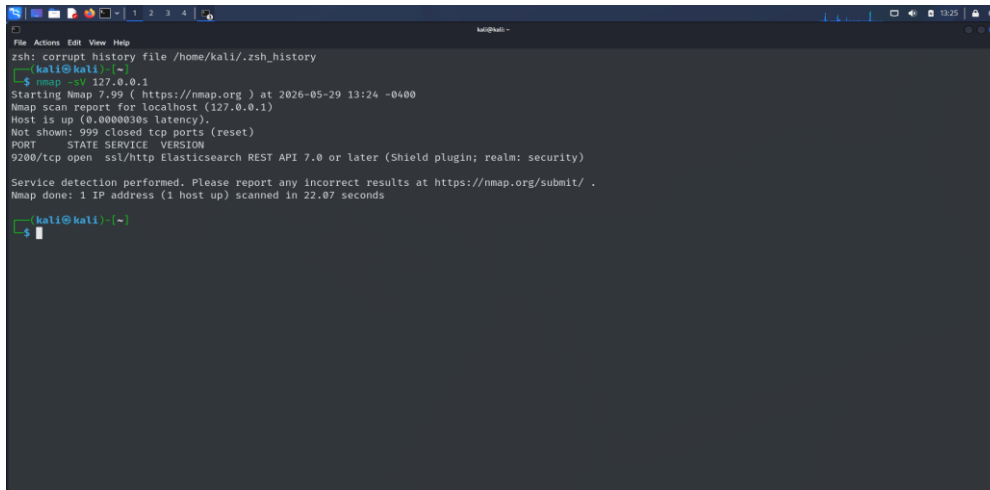
kali@kali:~$
kali@kali:~$ sshort -v
-# Sshort: de
Version 3.12.2.0
By Martin Roesch & The Sshort Team
http://sshort.org/contact#team
Copyright (C) 2014-2026 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using DAQ version 3.0.24
Using libcap version 1.10.6 (64-bit time_t, with TPACKET_V3)
Using LuaJIT version 2.1.1761786044
Using LZMA version 5-8-3
Using OpenSSL 3.6.1 27 Jan 2026
Using PCRE2 version 10.46 2025-08-27
Using ZLIB version 1.3.1

kali@kali:~$

```

Figure: Snort IDS verification on Kali Linux.

Network Scanning



```
File Actions Edit View Help
kali@kali ~$ zsh: corrupt history file /home/kali/.zsh_history
kali@kali ~$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-29 13:24 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
9200/tcp   open  ssl/http Elasticsearch REST API 7.0 or later (Shield plugin; realm: security)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.07 seconds

kali@kali ~$
```

Figure: Nmap service version scan identifying Elasticsearch service on port 9200.

5.5. Additional Security Concepts

CIA Triad

Confidentiality

Confidentiality ensures that sensitive information is accessible only to authorized users. Security controls such as encryption, authentication, and access control mechanisms help protect data from unauthorized disclosure.

Integrity

Integrity ensures that data remains accurate, complete, and unaltered. Mechanisms such as hashing, digital signatures, and access controls are used to maintain data integrity and prevent unauthorized modifications.

Availability

Availability ensures that systems, applications, and data are accessible to authorized users whenever needed. Techniques such as backups, redundancy, fault tolerance, and disaster recovery planning help maintain availability.

Threat, Vulnerability, and Risk

Threat

A threat is any potential event, action, or actor that can cause harm to an information system, network, or organization.

Vulnerability

A vulnerability is a weakness or flaw in a system, application, or process that can be exploited by a threat actor.

Risk

Risk is the likelihood that a threat will exploit a vulnerability and cause damage to an organization's assets, operations, or reputation.

Defense in Depth

Defense in Depth is a cybersecurity strategy that uses multiple layers of security controls to protect systems and data. These layers may include firewalls, intrusion detection systems, endpoint protection, access controls, and security monitoring solutions.

Zero Trust

Zero Trust is a security model based on the principle of "Never Trust, Always Verify." Every user, device, and application must be continuously authenticated, authorized, and validated before accessing organizational resources, regardless of whether they are inside or outside the network.

6. Challenges Faced and Solutions

Challenge 1

Difficulty understanding SIEM query syntax.

Solution:

Practiced KQL queries and reviewed SIEM documentation.

Challenge 2

Large volumes of logs made analysis difficult.

Solution:

Applied filters and dashboard visualizations to focus on relevant events.

Challenge 3

False-positive alerts were generated.

Solution:

Fine-tuned alert thresholds and improved rule conditions.

7. Learning Outcomes

After completing this task, I gained knowledge of:

- Security Operations Center (SOC) functions.
- Security monitoring techniques.
- Log collection and analysis.
- Incident response procedures.
- SIEM platforms and security tools.
- Security event documentation.
- Alert creation and dashboard monitoring.

8. Conclusion

This task provided practical exposure to Security Operations Center operations, security monitoring, log management, and incident response procedures. Various SOC concepts, security tools, and workflows were studied and implemented. Through log analysis, dashboard creation, and alert configuration, a better understanding of threat detection and security operations was achieved. The task successfully strengthened both theoretical and practical cybersecurity skills.

9. References

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4. <https://www.tenable.com/products/nessus>
5. <https://attack.mitre.org>
6. <https://www.wireshark.org/docs>
7. <https://www.elastic.co/guide>
8. <https://learn.microsoft.com/en-us/azure/sentinel/>